

Parallel Per-tile Activation with Linear Superposition of Thermal Response for Solving Arbitrary Power Pattern in 3DIC Thermal Simulation

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ABSTRACT

A large on-chip peak temperature and thermal gradient, caused by localized hotspots and cross-die thermal coupling, can severely impact transistor performance, stress, aging, electromigration (EM), voltage drops, and timing. Generating static and transient chip thermal profiles with traditional FEM/CFD methods have prohibitively expensive computational requirements, especially for three-dimensional integrated circuits (3DIC). Even though data driven based deep learning approaches can significantly improve the efficiency of obtaining accurate thermal profiles, large amounts of high-quality and diverse training data are required. Therefore, there is an urgent need for improving the data/training efficiency and generalizability of deep learning-based models for thermal applications. This investigation leverages the “linear superposition principle” for linear systems and divides a complex power pattern into “per-tile activation” scenarios. The predictions from machine learning (ML) model trained on per-tile activation data, which consists of power and thermal response pairs, are then linearly combined to generate the total thermal response for a given arbitrary power pattern. This novel approach can substantially reduce the infinite design space of the power patterns to a small finite design space, and it has been validated that it can significantly improve the generalizability and training efficiency of deep learning models. The proposed approach can be applied to static and transient chip thermal simulation for 3DIC.

CCS CONCEPTS

• Computing methodologies • Applied computing • Mathematics of computing

KEYWORDS

Chip thermal, Deep learning, Superposition principle, FEA, 3DIC

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1 Introduction

The detrimental impact of thermal issues on the performance and reliability of a chip is widely acknowledged [1][2][3]. Excessively high peak temperatures and large thermal gradient would cause issues such as reducing the efficiency and speed of transistors, incurring physical deformations like warping or micro-cracking, accelerating electromigration, affecting the chip's power supply stability, disrupting data processing or communication and so on [4][5].

Traditional FEA (Finite Element Analysis) based thermal analysis can be computationally demanding and time consuming due to the complexity of the model. Additionally, each distinct chip configuration within the extensive parameter space would require a new thermal analysis from scratch. Over the past few years, deep learning models have demonstrated considerable potential in expediting traditional FEA for thermal simulation. By embedding physical laws into deep learning models and utilizing Automatic Differentiation (AD) to calculate Partial Differential Equations (PDEs) derivatives, physics-informed machine learning (PINN) [6][7] have demonstrated effectiveness in tackling a variety of forward and inverse problems, including those related to heat transfer [8]. Benefiting from the implementation of physics constraints, these models require very few labeled data points for training. Nevertheless, when presented with a fresh problem configuration, the networks necessitate retraining, akin to the procedures observed in conventional FEA [9]. To alleviate the issue, a range of solutions have been introduced. For instance, instead of solving a single instance at training time, batches of cases are handled simultaneously during training in a physics-informed approach in [10]. Moreover, various data-driven methods [11][12][13][14][15], including mesh independent Neural Operator based approaches [16][17], have been proposed for solving complex physical systems and PDEs including heat transfer equations, demonstrating significant promise. Nevertheless, these models are trained using high-fidelity solutions derived from conventional PDE solvers, without necessitating any prior knowledge of the PDE. Thus, the availability of abundant, high-quality data is crucial and often becomes the bottleneck for data-driven based approaches. And acquiring adequate high-quality data which could uniformly cover

the entire parameters space of intricate problems, such as 3DIC, presents a formidable challenge. Thus, enhanced PINN and data-driven based frameworks have been suggested to address this challenge. For instance, inspired by PINN and DeepONet [16], a physics-aware operator learning framework [18] has been developed for thermal prediction. And a hybrid solution workflow [19], which integrates domain knowledge and data, have been suggested to augment prior techniques [11][12] to provide thermal solutions for 3DIC.

Despite aforementioned promising outcomes, few literatures directly tackle the issue with respect to the enormous parameter space involved in 3DIC, especially related to arbitrary and drastically varying power patterns. As a consequence, the training efficiency and generalization capability of the previously mentioned methods remain uncertain when encountering unseen scenarios, particularly with regard to arbitrarily complex power patterns. To address the issues mentioned above, this paper introduces a novel method for handling arbitrary power patterns in 3DIC, thereby significantly improving the generalization capability of existing techniques. The unique contributions of this investigation are as follows:

1. Introducing a novel per-tile activation-based data generation approach, which eliminates the need to enumerate all potential combinations of power patterns for obtaining training data covering the entire power pattern space. The resulting reduction in the power pattern parameter space will facilitate efficient and effective training of deep learning models and ensures high accuracy with a significant enhancement in their generalization capability.
2. A complete solution workflow, which leverages above per-tile activation based method and thermal linear superposition [20][21], is proposed for solving arbitrary power pattern in 3DIC thermal simulation.
3. The proposed solution workflow can be readily adapted to address transient thermal problems.
4. A comprehensive framework, which integrates Latin Hypercube Sampling (LHS), power-temperature (P-T) based solution refinement iteration loop, Decay Curve Characterization [19], and Transient Response Curves (local/intermediate/global) [22] for solving steady state and transient thermal for 3DIC under different configurations is presented.

2 Chip Thermal Analysis

2.1 Modelling for Chip Thermal Analysis

Chip Thermal Analysis is a vital process that delves into the intricate temperature distribution and thermal behavior of integrated circuits (ICs). By scrutinizing these aspects, it assists in hotspots identification, design optimization, reliability assessment and so on [11][23]. Chip thermal analysis encompasses a diverse array of components, layers of distinct materials and complicated structures.

It has been verified [3], [10], [12], [19] that simplified N-layer model (Typically, N=3 or 4) can serve as a reasonable candidate for chip modelling in terms of reasonability, accuracy, efficiency in batch data generation. A schematic of a 3-layer model and a 4-layer model [24] is provided in Figure 1. It is worth noting that both models contain an interconnect layer, an insulation layer and

a Si substrate layer, except that the 4-layer model contains an extra layer of backside power delivery network (BSPDN). This paper focuses on the 3-layer model and this model is also used for stacked two-die scenarios, as illustrated by a stacked two-die example in Figure 1 (c).

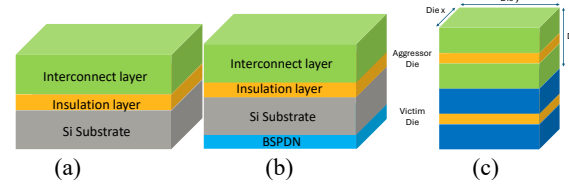


Figure 1. Schematic of N-layer models (a) 3-layer model, (b) 4-layer model, (c) two stacked dies.

In this paper, the system parameters defining the 3-layer model are provided in Table 1, which are obtained based on empirical experience from domain experts. These parameters and their corresponding values will also be used to guide the data generation for deep learning models.

Table 1. System parameters of the 3-layer model

System parameters	Values
Per tile Power (mW)	50
Heat Transfer Coefficient (mW/μm ² •K)	1e ⁻⁸ -1e ⁻⁶
Die x / y length (μm)	4000
Die z height (μm)	50 - 200
Dielectric/interconnection layer thickness (μm)	5-10
Insulation layer thickness (μm)	0.05
Tile size (μm)	200
Si substrate thermal conductivity (mW/μm•K)	0.1305
Insulation layer thermal conductivity (mW/μm•K)	1.38e ⁻³
Dielectric / interconnection layer thermal conductivity (mW/μm•K)	1.38e ⁻³

2.2 Physics for Chip Thermal Analysis

Given the modeling mentioned above, the steady state temperature distribution is governed by the following thermal diffusion Equation (1):

$$\frac{\partial}{\partial x} \left(K_x \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(K_y \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(K_z \frac{\partial T}{\partial z} \right) = Q \quad (1)$$

Where T is the temperature distribution of the chip, K_x , K_y and K_z is the thermal conductivity in the x, y and z direction respectively and Q represents the heat sources. It is worth noting that this fundamental equation characterizes a linear system, which significantly contributes to the implementation of thermal linear superposition. Additionally, based on empirical experience, under uniform and distributed heat transfer coefficients (HTC)—the most typical boundary conditions for chip applications, linear superposition of thermal responses is accurate and effective.

3 Parallel Per-tile Activation with Linear Superposition of Thermal Responses

3.1 Per-tile Activation

One of the major challenges presented in chip thermal analysis is introduced by the source term Q in Eq. (1). The heat source term originates from numerous cells within the functional blocks, typically modeled as rectangular regions, of the chip. These functional blocks may vary in size and exhibit drastically different power values.

To effectively represent and modularize the functional blocks across the chip, the concept of “tile” has been introduced [19]. These tiles, which can be considered as subdivisions or unit areas of the chip, enable the organization and management of complex designs more efficiently. Practically, each tile can function independently (exhibit distinct power values) or in coordination with other tiles. For illustrative purposes, a simplified tile-based power pattern is shown in Figure 2 (a), which consists of 4 tiles with distinct power values. The corresponding per-tile activation of this power pattern is illustrated in Figure 2 (b), where the original full power pattern scenario is decomposed into four single tile activation cases. It is worth mentioning that in practical applications, as described in Table 1, for example, a tile size of $200\mu\text{m} \times 200\mu\text{m}$ is typically chosen for a chip measuring $4\text{mm} \times 4\text{mm}$ to balance accuracy and computational cost.

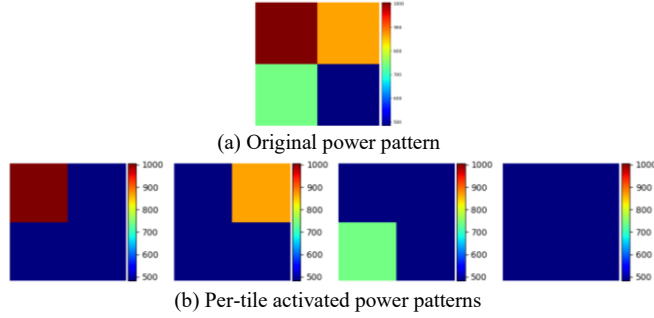


Figure 2. Schematic of 2x2 tile-based power patterns

3.2 Linear Superposition of Thermal Responses

According to the superposition property of linear systems [25][26], the net response caused by two or more stimuli is the sum of the responses that would have been caused by each stimulus individually. Thus, regarding thermal linear superposition, the assumption is that the overall impact of activating all components/tiles simultaneously could be expressed as a combination of individual thermal effects resulting from activating each component/tile independently with the corresponding tile power.

To validate above assumption, the power pattern described in Figure 2 (a) is investigated. Firstly, the temperature distribution for the power pattern in Figure 2 (a) is calculated using a commercial 3DIC thermal analysis software [27], as shown in Figure 3 (a). Then the numerical solutions for the per-tile activation-based power patterns in Figure 2 (b) are obtained and presented in Figure 3 (b). Finally, a comparison between the ground truth solution in Figure 3 (a) and the linear superposition of the 4 solutions in Figure 3 (b) is shown in Figure 4.

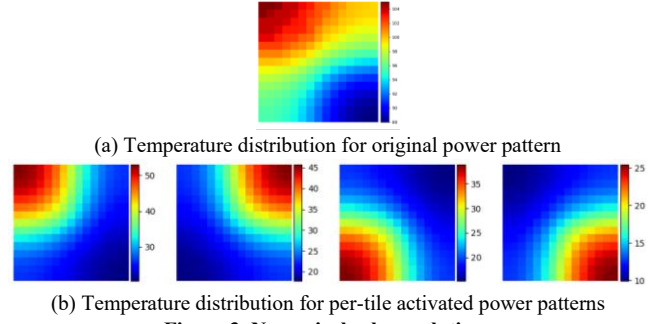


Figure 3. Numerical solver solutions

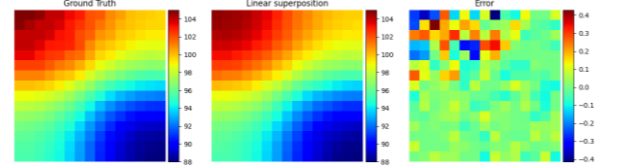


Figure 4. Comparison between solutions of original power pattern and linear superposition of per-tile activated power patterns.

According to the error plot in Figure 4, the mean average percentage error (MAPE) is 0.0007 (0.07%). It is assumed that one of the major factors causing this discrepancy might be some inherent numerical solver settings such as mesh settings (e.g., the power patterns’ boundary and mesh boundary are not perfectly aligned). Despite that, the performance of linear superposition is reasonable and acceptable. Additionally, the identical method can be readily extended to scenarios involving stacked two-die configurations, as depicted in Figure 1 (c). In this scenario, per-tile activation will be executed on every tile of both the top and bottom dies, and the resulting thermal responses will be aggregated to obtain the ultimate thermal profile.

4 Per-tile Activation Based Data Generation and Solution Workflow

The proposed solution workflow is illustrated in Figure 5. The deep learning model utilized in this investigation is our previously developed machine learning-based thermal solver, which has been validated in prior research [12][19]. However, it is important to note that this solution workflow is not limited to any specific deep learning models or machine learning-based solvers. Thus, the module labeled “ML-based thermal solver” in Figure 5 can be replaced with various state-of-the-art models, such as those mentioned in the Introduction section.

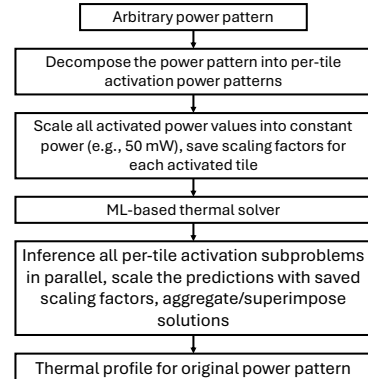
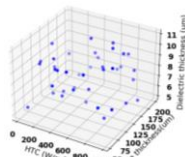


Figure 5. Solution workflow

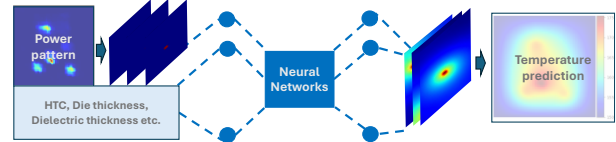
4.1 Training Data Generation

The enormous design space associated with power patterns has been widely recognized as a longstanding challenge, presenting significant obstacles for data-driven chip thermal analysis. For example, for the same example mentioned in previous section, the power pattern for a 4mm x 4mm chip, 200 μ m x 200 μ m tile size would consists of 20 x 20 power tiles. Considering that each of the previously mentioned 20x20 tiles could exhibit any power value within a certain range (e.g., 0 to 330mW), the total combinations of the power patterns could theoretically reach N^{400} , where N represents the number of possible values for each tile. Therefore, it is impractical to generate a comprehensive set of power patterns to fully cover the potential power pattern space. However, by leveraging the linear property of thermal responses and only focusing on 50 mW per-tile activation scenarios (a reasonably chosen value of power for a single tile), the previously infinite power pattern space can be effectively covered using just 400 per-tile activation patterns. This crucial step considerably narrows the infinite power pattern space to a manageable finite space. Moreover, by additionally capitalizing symmetry property of these 400 per-tile activation patterns with a constant power of 50mW, the quantity could be further decreased from 400 to 100 for the example chip.

Therefore, for the configurations mentioned in Table 1, a 3-layer chip model, as depicted by Figure 1 (a), is modeled using a commercial 3DIC thermal analysis software for data generation. Besides, since the dimension of the chip along the z axis is considerably smaller than its dimensions along the x and y axis, this study primarily focuses on temperature variations across the x-y plane. For illustration, this paper focuses on a 4mm x 4mm chip with 200 μ m x 200 μ m tile size, which requires only 400 50mW per-tile activation power patterns. Second, for validation on finer power patterns (4mm x 4mm chip, 100 μ m x 100 μ m tile size), a dataset containing 1600 power patterns has been generated as well. Third, to investigate the associated system parameters, three major factors influencing the chip temperature are varied for creating another dataset. These factors include the heat transfer coefficient, ranging from $1e^{-8}$ to $1e^{-6}$ mW/ μ m²•K; the thickness of the die, varying from 50 μ m to 200 μ m; and the thickness of dielectric layer, ranging from 5 to 10 μ m. To ensure thorough coverage of the parameter space and to capture the realistic variability in practical applications, Latin Hypercube Sampling is utilized to sample 40 sets of system parameters (shown in Figure 6) and generate various numerical cases. In total, there are 16000 (which can be further reduced to 4000 by leveraging symmetry properties) cases generated for experimental validation on both power pattern variation and system parameter variations.

**Figure 6. Latin Hypercube Sampling of system parameters**

With the generated data, the machine learning-based thermal solver is trained. The specifics regarding the machine learning-based thermal solver and its related settings are consistent with those outlined in [12][19]. To reiterate, since this workflow is independent of the specific deep learning models used, most state-of-the-art models, such as those mentioned in the Introduction section, can be employed. Because the substantial reduction in the power pattern space significantly enhances both data generation efficiency and the accuracy of trained deep learning models. A schematic of deep learning model prediction is displayed in Figure 7.

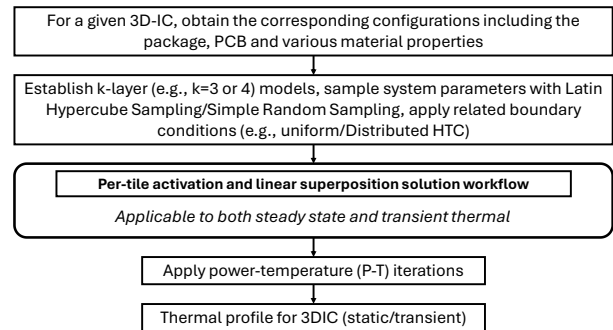
**Figure 7. Schematic of deep learning model**

4.2 Solution Workflow for Arbitrary Power Patterns

At inference stage, as shown in Figure 5, given any arbitrary power pattern, it will first be decomposed into per-tile activation power patterns. The power value for each tile will be linearly scaled with respect to 50 mW and the corresponding scaling factors will be recorded for subsequent temperature scaling. The trained deep learning model will perform inference on all per-tile (50 mW) activation subproblems in parallel to obtain the corresponding thermal profiles. These thermal profiles will then be scaled by the scaling factors saved earlier and subsequently aggregated through linear superposition to produce the final output of the solution workflow.

4.3 Comprehensive Framework for Steady/Transient 3DIC

Built upon the above mentioned solution workflow, the comprehensive framework for tackling 3DIC is proposed, as illustrated by the schematic in Figure 8.

**Figure 8. Schematic of the comprehensive framework for tackling 3DIC**

Therefore, given a System on a Chip (SoC), its configurations such as the package, PCB and material properties will be extracted. Then a few k-layer ($k = 3$ or 4) models will be generated using a commercial 3DIC thermal analysis software. Per-tile activation based power patterns, together with sampled

system parameters (e.g., using Latin Hypercube Sampling) creates the datasets for training the model.

1) Extension for 3DIC

The above methodology can be easily extended for 3DIC by such that the per-tile activation ML model predicts the temperature in the x, y, z directions effectively accounting for vertically stacked dies and the associated thermal coupling.

2) Fine-grained thermal analysis

For scenarios where fine-grained (e.g., 10 μ m tile size) thermal solution is required, this solution workflow can be easily integrated with previous workflows [19].

3) Extension for transient thermal

The same per-tile activation approach can also be extended to address transient thermal analysis such that the ML model predicts transient thermal profiles at the victim tiles. The transient thermal linear superposition proposed in [3] can be directly applied to compose the effective transient thermal response at tile locations.

4) P-T iteration

Leakage power has an exponential dependency on the temperature. To account for this effect the above methodology performs Power-Temperature iterations based on the provided relationship to arrive at a converged thermal solution after the linear superposition step.

5 Experiments, Results and Discussions

To validate the effectiveness of the proposed approach, testing has been performed under various scenarios: (1) arbitrary power patterns generated by random permutation of power values (2) power patterns observed in real-world applications (3) power patterns on different floor plans (4) arbitrary power patterns with a fine resolution of 100 μ m x 100 μ m tile size; and (5) arbitrary power patterns with varying system parameters. The golden results for comparison have been generated using a commercial 3DIC thermal analysis software.

5.1 Validation on Randomly Generated Power Patterns

Power values within a range of 0 to 330mW, as derived from empirical experience, are used to randomly generate 5000 distinct power patterns for evaluation. Examples of two power patterns are depicted in the first column of Figure 9. The subplots in the second, third and fourth columns represent the ground truth solution, predictions from deep learning model and the element-wise absolute error, respectively. The total MAPE over all 5000 cases is calculated to be 0.00069, demonstrating remarkable accuracy. It is important to note that the training dataset included only 400 per-tile cases.

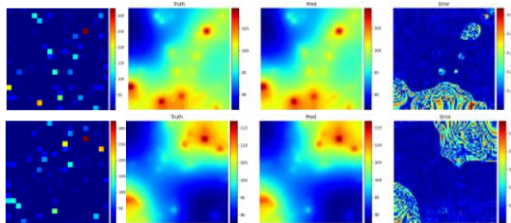


Figure 9. Random artificial power patterns

5.2 Validation on Real-world Power Patterns

Three power patterns observed in distinct real-world applications are extracted for analysis and depicted in Figure 10 (a), (b) and (c). The prediction results of power pattern (a), (b) and (c) are displayed in the first, second and third rows of Figure 11, respectively. In Figure 11, the first column of each subplot illustrates the ground truth solution, the second column presents the predictions from the deep learning model, and the last column demonstrates the corresponding element-wise absolute error. Furthermore, the MAPE for these three testing scenarios are 0.0007, 0.0003 and 0.0005 respectively, indicating desirable results.

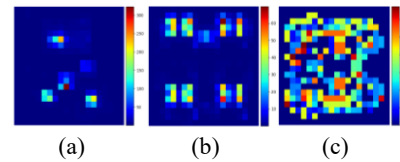


Figure 10. Examples of arbitrary power patterns

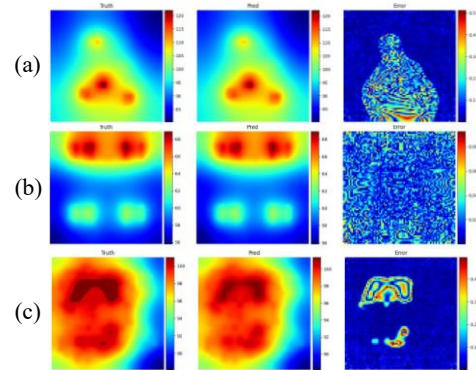


Figure 11. Predictions of the trained model on three real-world power patterns (1st column: ground truth, 2nd column: model prediction, 3rd column: element-wise absolute error)

5.3 Validation on Different Floorplans

An important use case of the proposed thermal analysis methodology is early thermal analysis at floorplan stage. Figure 12 shows the following study: (a) and (b) have identical floorplans (4mmx4mm chip) but different powers assigned to the blocks and (c) has a different floorplan but same powers assigned to the blocks as in (b). The multitude of combinations of power patterns and floorplans further adds significant complexity to chip thermal simulation. The experiments show that all these different combinations can lead to vastly different hotspots and thermal profiles.

To validate the efficacy of the proposed solution workflow, the same model described in Section V.A and Section V.B is used to predict temperatures for the power patterns labeled (a), (b) and (c) in Figure 12 bottom row. The corresponding predictions are presented in the first, second and third row of Figure 13, with mean average percentage error of 0.0012, 0.0017 and 0.0012 for each, respectively.

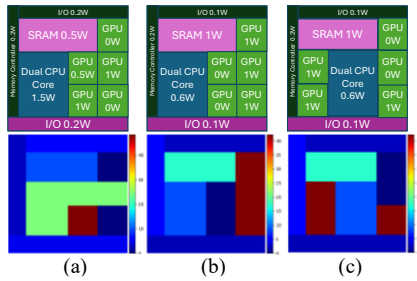


Figure 12. Examples of floor plans (top) and corresponding power patterns (bottom)

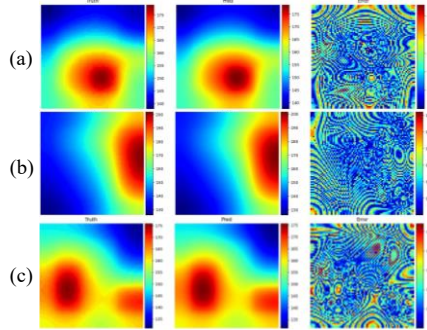


Figure 13. Predictions of the trained model on various power pattern and floorplan combinations (1st column: ground truth, 2nd column: model prediction, 3rd column: element-wise error)

5.4 Validation on Power Pattern with 100 μm x 100 μm tile size

To verify the effectiveness of the proposed approach with even more fine-grained power patterns, per-tile activation training cases for a tile size of 100 μm x 100 μm are generated for the 4mm x 4mm chip. A total of 1000 random testing cases are generated using a numerical solver and two example power patterns are depicted in the first column of Figure 14. The second, third and fourth column of Figure 14 display the ground truth solution, predictions from the trained model and the corresponding element-wise absolute error respectively. The calculated MAPE for the trained model's prediction across all 1000 testing cases is 0.001456, indicating satisfactory performance on fine-grained power patterns.

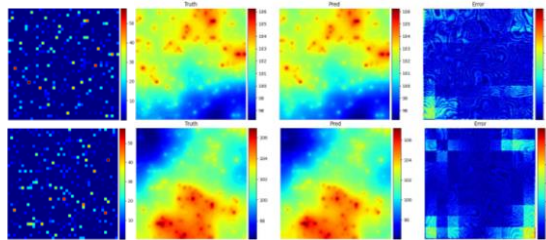


Figure 14. Two examples of 100 μm x 100 μm testing power patterns.

5.5 Validation on Different Power Patterns with Different System Parameters

In practical applications, besides the drastically varying power patterns, various system parameters, as described in Table 1, may also affect the temperature profile of the chip. Based on empirical experience, three key factors influencing chip temperature are

identified and investigated. These factors consist of the heat transfer coefficient, die thickness and dielectric layer thickness. The process of parameter sampling and data generation pertaining to these factors has been elaborated upon in Section **Error! Reference source not found.** For validation purposes, power patterns are randomly generated, and system parameters are randomly selected from the parameter space mentioned in Section **Error! Reference source not found.** Examples of the testing cases are illustrated in Figure 15. The corresponding system parameters for the top test case are heat transfer coefficient of $6.86\text{e-}7 \text{ mW}/\mu\text{m}^2\cdot\text{K}$, die thickness of 71 μm and dielectric layer thickness of 9 μm . For the bottom case, the parameters include heat transfer coefficient of $4.31\text{e-}7 \text{ mW}/\mu\text{m}^2\cdot\text{K}$, die thickness of 78 μm , and dielectric layer thickness of 8 μm . The MAPE of the trained model across 1000 test cases is calculated to be 0.013, which is excellent considering the selection of only 40 points from the entire parameter space, while also ensuring generalizability to arbitrary power patterns. It is believed that finer sampling of system parameters, beyond what is depicted in Figure 6 (40 points), could further improve accuracy.

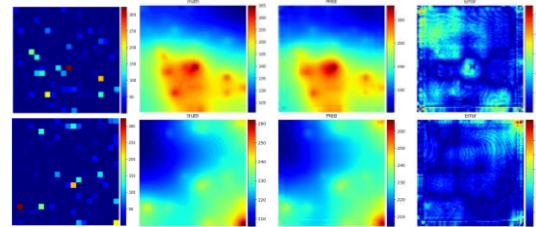


Figure 15. Example test cases with random power patterns and system parameters.

Table 2 shows the runtime comparison between the FEA based thermal analysis and proposed ML-based solution workflow on an Intel Xeon Gold 6226R CPU. The runtime improvement for a 4 mm x 4 mm die with 200 μm x 200 μm tile size is 272.4X. As the die size increases and package and PCB becomes more complex, the FEA based analysis becomes prohibitively expensive for fine grained thermal analysis. By combining the proposed solution workflow with previously published decay curve approach [19], it will provide even more speed up.

Table 2 Runtime improvement using our proposed ML based thermal analysis

Test Case	FEA Based	Ours	Speed-up
4 mmx4 mm, 200 μm x 200 μm	1m19s	0.29s	272.4X

6 Conclusions

This paper proposed a novel solution framework that leverages the “linear superposition principle” for chip thermal simulation. By substantially reducing the infinite design space of power patterns to a small finite design space, it greatly improves the generalizability and training efficiency of deep learning models for thermal profile prediction. The effectiveness and accuracy of this approach have been validated against a commercial 3DIC thermal analysis software across various scenarios including random artificial and realistic power patterns, different floor plans, distinct fine-grained power patterns and variations on both power pattern and system parameters. The proposed approach can be applied to static and transient chip thermal simulation for 3DIC.

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